Also $\forall$ distributes over $\wedge$ and $\exists$ over $\vee$.

$\forall$ does not distribute over $\vee$ and $\exists$ over $\wedge$. Also if $F$ has a variable $x$ and $G$ does not contain $x$, then

$$\left.\begin{array}{ll} (Qx)\, F(x) \vee G \longleftrightarrow Q(x)\,(F(x) \vee G) & (3) \\ (Qx)\, F(x) \wedge G \longleftrightarrow Q(x)\,(F(x) \wedge G) & (4) \end{array}\right\} \text{(Ex-1.4, Prob 48 and 49)}$$

We see that if $F_1$ and $F_2$ have variable $x$,

$$\forall x F_1(x) \vee \forall x F_2(x) \neq \forall x (F_1(x) \vee F_2(x)) \quad \text{(Ex-1.4, Prob 52)}$$

But in $F_2(x)$ we can rename the variable $x$ as $z$ and get $\forall x F_1(x) \vee \forall z F_2(z)$ which can be brought to the form $\forall x \forall z (F_1(x) \vee F_2(z))$. Note that $F_2$ does not contain $x$ and $F_1$ does not contain $z$.

Similarly, $\exists x F_1(x) \wedge \exists x F_2(x)$ can be brought to the following form by renaming of variable $x$ as $z$ in $F_2$.

$$\begin{aligned} &\exists x F_1(x) \wedge \exists x F_2(x) \\ &= \exists x F_1(x) \wedge \exists z F_2(z) \\ &= \exists x \exists z (F_1(x) \wedge F_2(z)) \end{aligned}$$

## Conversion of a formula in first order logic to prenex normal form

Step 1: Replace $\longleftrightarrow$ and $\longrightarrow$ using $\wedge, \vee, \neg$.

Step 2: Use Double negation and De Morgan's Laws repeatedly and the laws (1) and (2).

Step 3: Rename variables if necessary.

Step 4: Use rules (3) and (4) to bring the quantifiers to the left.

**Example 8:** Transform the formula into prenex normal form

$$\begin{aligned} &\forall x P(x) \longrightarrow \exists x Q(x) \\ &\equiv \neg \forall x P(x) \vee \exists x Q(x) && \text{(Conditional-Disjunction Equivalence)} \\ &\equiv \exists x \neg P(x) \vee \exists x Q(x) && (1) \\ &\equiv \exists x (\neg P(x) \vee Q(x)) && (\exists \text{ distributes over } \vee) \end{aligned}$$

**Example 9:** Obtain prenex normal form for the formula

$$(\forall x)(\forall y)((\exists z)(P(x,z) \wedge P(y,z)) \longrightarrow (\exists u)\, Q(x,y,u))$$

$$\equiv (\forall x)(\forall y)(\neg(\exists z)(P(x,z) \wedge P(y,z)) \vee (\exists u)\, Q(x,y,u)) \quad \text{(conditional-disjunction equivalence)}$$